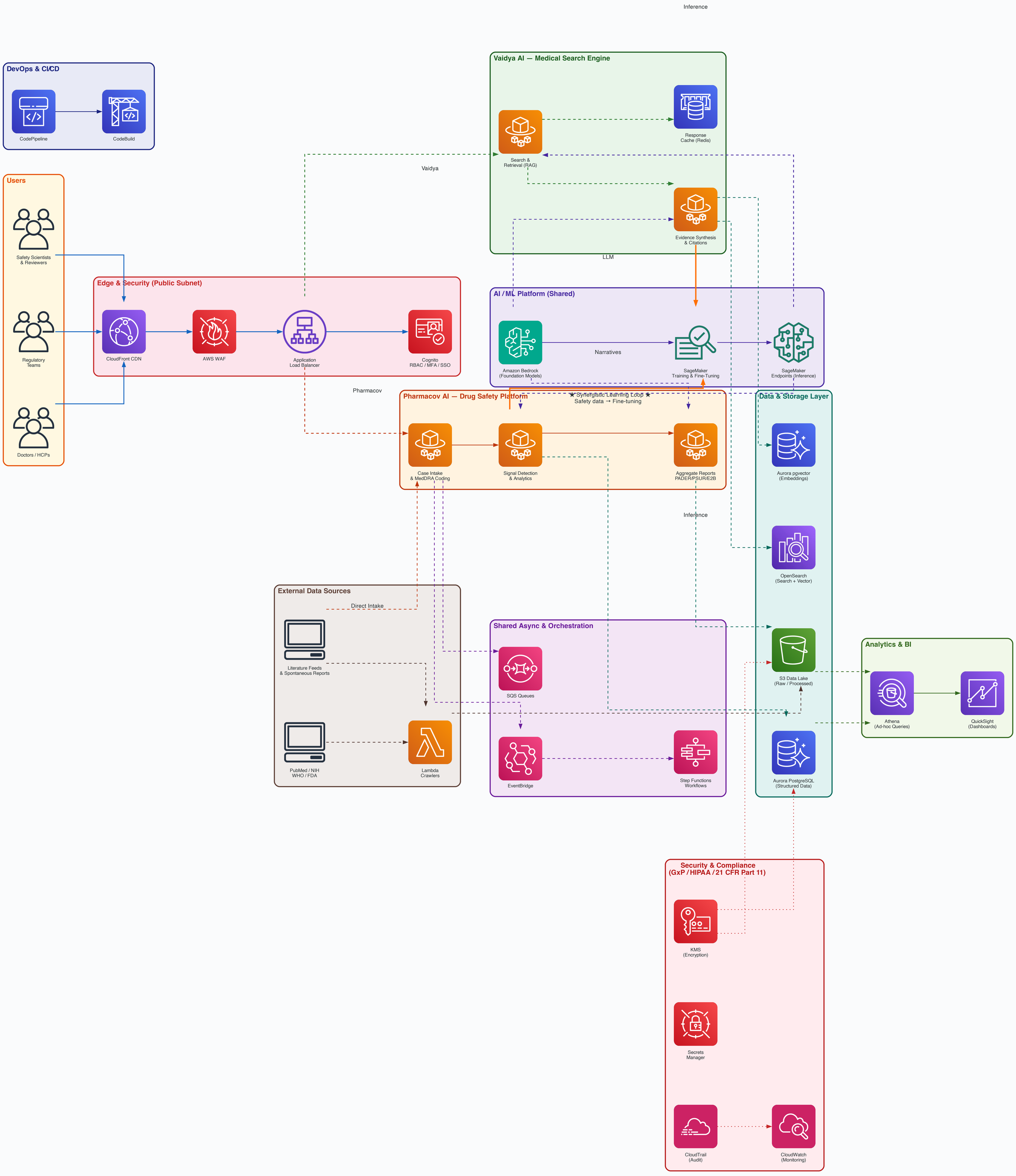
Inference
DevOps & CI/CD
CodePipeline
CodeBuild
Vaidya AI — Medical Search Engine
Search & Retrieval (RAG)
Response Cache (Redis)
Vaidya
Evidence Synthesis & Citations
LLM
Users
Safety Scientists & Reviewers
Regulatory Teams
Doctors / HCPs
Edge & Security (Public Subnet)
CloudFront CDN
AWS WAF
Application Load Balancer
Cognito RBAC / MFA / SSO
AI / ML Platform (Shared)
Amazon Bedrock (Foundation Models)
Narratives
SageMaker Training & Fine-Tuning
SageMaker Endpoints (Inference)
Pharmacov
Pharmacov AI — Drug Safety Platform
★ Synergistic Learning Loop ★
Safety data → Fine-tuning
Case Intake & MedDRA Coding
Signal Detection & Analytics
Aggregate Reports PADER/PSUR/E2B
Data & Storage Layer
Aurora pgvector (Embeddings)
Inference
OpenSearch (Search + Vector)
External Data Sources
Direct Intake
Literature Feeds & Spontaneous Reports
PubMed / NIH WHO / FDA
Lambda Crawlers
Shared Async & Orchestration
SQS Queues
EventBridge
Step Functions Workflows
S3 Data Lake (Raw / Processed)
Aurora PostgreSQL (Structured Data)
Analytics & BI
Athena (Ad-hoc Queries)
QuickSight (Dashboards)
Security & Compliance (GxP / HIPAA / 21 CFR Part 11)
KMS (Encryption)
Secrets Manager
CloudTrail (Audit)
CloudWatch (Monitoring)